

POPULATION HEALTH BRIEF

Top 10 Highest-Risk U.S. Counties | CDC PLACES 2023 + SVI 2022 | K-Means Risk Stratification (K = 4)

Background

County-level CDC PLACES health estimates (2019–2023) were merged with the CDC/ATSDR Social Vulnerability Index (SVI) on FIPS code and standardized via z-score scaling. K-Means clustering (K = 4) segmented all ~3,000 U.S. counties into four risk tiers: **Low**, **Moderate**, **High**, and **Critical**. Counties in the **Critical** tier ($n = 238$, 10% of all counties) were ranked by a composite score averaging normalized diabetes prevalence, hypertension prevalence, and overall SVI. The 10 highest composite-scoring counties are presented below.

Critical vs. National Average (2023)

Indicator	Critical	National
Diabetes prevalence	15.8%	11.1%
Hypertension prevalence	42.9%	33.9%
Physical inactivity	35.8%	27.2%
Obesity prevalence	44.1%	38.1%
Current smoking	21.1%	16.5%
Food insecurity	30.5%	17.3%
Lack of transportation	15.6%	9.5%
Housing insecurity	23.1%	14.0%
SVI (overall, 0–1)	0.90	0.48

Top 10 Highest-Risk Counties

Rank	County	State	Pop.	Diabetes %	Hypert. %	SVI	Phys. Inact. %	Obesity %	Smoking %
1	Holmes	Mississippi	15,777	21.0	53.1	0.992	44.4	51.5	23.5
2	Humphreys	Mississippi	7,216	20.6	52.4	0.979	46.2	53.0	24.1
3	Greene	Alabama	7,341	21.1	52.4	0.916	42.0	53.4	23.8
4	Bullock	Alabama	9,897	21.3	52.4	0.899	42.7	51.0	24.9
5	E. Carroll	Louisiana	6,829	21.3	52.5	0.878	44.1	51.7	29.1
6	Perry	Alabama	7,738	20.5	51.9	0.921	40.3	54.0	23.3
7	Sharkey	Mississippi	3,336	19.6	51.2	0.984	44.3	50.5	23.5
8	Tunica	Mississippi	9,234	19.9	51.2	0.960	44.8	50.8	23.6
9	Coahoma	Mississippi	20,077	19.5	50.7	0.990	42.1	47.1	22.0
10	Madison	Louisiana	9,246	19.6	49.6	1.000	42.5	50.1	26.2

Recommended Interventions

- **Food access:** Expand SNAP outreach and mobile food pantry programs; prioritize counties with food insecurity >40%.
- **Transportation:** Fund community health worker home-visit models and non-emergency medical transport in counties where transportation access gaps exceed 15%.
- **Preventive care:** Deploy federally qualified health center (FQHC) satellites targeting hypertension screening and diabetes management in Critical tier counties.
- **Physical activity:** Partner with schools and faith commu-

Key Findings

- The top 10 highest-risk counties are concentrated in the **Deep South** (Mississippi 5, Alabama 3, Louisiana 2), reflecting persistent regional disparities in chronic disease burden.
- All 10 counties have diabetes rates $\geq 19.5\%$, nearly **double** the national average of 11.1%.
- Hypertension exceeds **49%** in every top-10 county; Holmes County, MS reaches **53.1%**.
- SVI scores cluster near the ceiling (≥ 0.88), signaling extreme concurrent vulnerability across socioeconomic, housing, and transportation domains.
- Physical inactivity ($r = 0.87$), lack of transportation ($r = 0.91$), and food insecurity ($r = 0.94$) are the strongest SDOH correlates of county-level diabetes in 2023 (Pearson r , $n = 2,299$ counties).

nities for low-cost activity programs given >40% physical inactivity rates.

Data & Methods Notes

Data: CDC PLACES 2023 release (data year 2023); CDC/ATSDR SVI 2022. Composite risk score = mean of min–max normalized diabetes prevalence, hypertension prevalence, and overall SVI (RPL_THEMES). K-Means validated via elbow and silhouette analysis; K = 4 selected. Age-adjusted prevalence used where available; crude otherwise. 657 counties excluded from clustering due to missing values in one or more clustering features (complete-case analysis; 2,956 in panel → 2,299 clustered).